

Math 675 Spring 2026: Project 2 Notes Due Thursday, May 14th, 2026
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Suggested Structure for Final Report

- **Introduction:** Problem, motivation, and context
- **Background:** Definitions and setup
- **Model Development:** Derivation and assumptions
- **Analysis:** Mathematical results and explanations
- **Discussion:** Interpretation, limitations, and data considerations
- **Conclusion:** Summary and extensions
- **References:** Literature and sources

Introduction

Background

Model Development

Analysis

Discussion

Conclusions

Reflection (1 Page)

- Include a short reflection addressing:
- What feedback did you receive on your poster?
- What did you improve in the presentation and paper?
- What mathematical aspects did you deepen?
- What did you learn about communicating applied math?

References (1 Page)

General Links for Sources about FEA

- “Why is a Triangular Element Stiffer?” *EnterFEA*.
<https://enterfea.com/why-is-a-triangular-element-stiffer/>

- “The Importance of Mesh Convergence.” *NAFEMS Knowledge Base*.
<https://www.nafems.org/publications/knowledge-base/the-importance-of-mesh-convergence-par>
- “What is Finite Element Analysis and How Does It Really Work?” *Fidelis FEA*.
<https://www.fidelisfea.com/post/what-is-finite-element-analysis-and-how-does-it-really-w>
- “What Is FEA — Finite Element Analysis?” *SimScale Documentation*.
<https://www.simscale.com/docs/simwiki/fea-finite-element-analysis/what-is-fea-finite-elem>
- “What Is Finite Element Analysis?” *MathWorks Discovery*.
<https://www.mathworks.com/discovery/finite-element-analysis.html>
- “Solving Partial Differential Equations with Finite Elements.” *Wolfram Reference*.
<https://reference.wolfram.com/language/FEMDocumentation/tutorial/SolvingPDEwithFEM.html>

Links for Thermal Property Values Estimator Infomation and Code

- “Material thermal properties estimation via a one-dimensional transient convection model.” *Applied Thermal Engineering*, ScienceDirect.
<https://doi.org/10.1016/j.applthermaleng.2020.116230>
- “Data on the Validation to Determine the Material Thermal Properties Estimation.” *Data in Brief*, ScienceDirect.
<https://doi.org/10.1016/j.dib.2021.107577>
- “Python code for 1D k and h value Estimator.” *Zenodo Repository*.
<https://zenodo.org/records/5683312>
- “oneDkhEstimator Required Dataset.” *Mendeley Data*.
<https://data.mendeley.com/datasets/sbcf36976g/1>